

**[bug] Measurement handling in `convert\_qiskit\_to\_braket\_circuit` sometimes**

<https://github.com/qiskit-community/qiskit-braket-provider/issues/132>

ROW 28

# [bug] Measurement handling in `convert_qiskit_to_braket_circuit` sometimes fails #132

New issue



Closed

#135



laurencap opened on Nov 17, 2023 · edited by laurencap

Edits · ...

This issue is the result of some investigation following this bug report: [#128](#)

## Steps to reproduce the problem

```
from qiskit.primitives import BackendSampler

from qiskit_algorithms.minimum_eigensolvers import QAOA
from qiskit_algorithms.optimizers import COBYLA
from qiskit_optimization.algorithms import MinimumEigenOptimizer
from qiskit_optimization.problems import QuadraticProgram
from qiskit_optimization.translators import from_docplex_mp

from qiskit_braket_provider import AWSBraketProvider

from docplex.mp.model import Model

mdl = Model("docplex model")
x0 = mdl.binary_var("x0")
x1 = mdl.binary_var("x1")
x2 = mdl.binary_var("x2")

mdl.minimize(-30*x0**2 + 14*x0*x1 + 14*x0*x2 - 35*x1**2 + 14*x1*x2 - 29*x2**2)

mod = from_docplex_mp(mdl)

sampler = BackendSampler(
    # backend=AWSBraketProvider().get_backend("Aspen-M-3"),
    backend=BraketLocalBackend(),
    skip_transpilation=False,
    options={"shots": 100}
)

qaoa = QAOA(sampler=sampler, optimizer=COBYLA())
qaoa_solver = MinimumEigenOptimizer(qaoa)
result = qaoa_solver.solve(mod)
```

## What is the current behavior?

```
ValueError: Observable's qubit count 1 and the size of the target qubit set QubitSet([Qubit(1), Qubit(0)])
must be equal
```

## What is the expected behavior?

Success

## Proposed solution

I believe that the existing method to convert Qiskit measurements to Braket result types is misusing the Braket API. This might fix the problem:

<https://github.com/qiskit-community/qiskit-braket-provider/compare/main...laurencap:qiskit-braket-provider:bugfix/convert-qiskit-to-braket-measurements?expand=1>



laurencap added **bug** on Nov 17, 2023

### Assignees

No one assigned

### Labels

**bug**

### Type

No type

### Projects

No projects

### Milestone

No milestone

### Relationships

None yet

### Development

**fix a misused 2-qubit sample result type**  
qiskit-community/qiskit-braket-provider

### Notifications

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### Participants



 **laurencap** added **bug** on Nov 17, 2023

 **laurencap** mentioned this on Nov 17, 2023

🔗 [Issues in running QAOA algorithm against Rigetti backend #128](#)

 **jcaskula-aws** mentioned this on Dec 29, 2023

🔗 [fix a misused 2-qubit sample result type #135](#)



jcaskula-aws on Dec 29, 2023 · edited by jcaskula-aws

Edits ▾

Collaborator



The fix is good and the bug is pretty bad. I shamelessly borrowed the fix and went further by renaming most of the variables in the function. Please have a look at [#135](#).



🔗  **ajberdy** closed this as **completed** in [#135](#) on Jan 11, 2024

# fix a misused 2-qubit sample result type #135

<> Code

Merged ajberdy merged 7 commits into qiskit-community:main from jcjaskula-aws:fix\_measure\_translation on Jan 11, 2024

Conversation 4 Commits 7 Checks 7 Files changed 2 +73 -25



jcjaskula-aws commented on Dec 29, 2023 · edited

Collaborator

## Summary

Measure instructions were wrongly translated. We now construct a 1-qubit sampling instruction.

## Details and comments

We were translating a 1-qubit measure instruction into a 2-qubit ZZ observable sampling instruction (using the classical register index for the second qubit). This PR uses the right instruction and renames most of the variables in the function to be clearer.

Fixes #132.



## Reviewers

ajberdy



rmshaffer



## Assignees

No one assigned

## Labels

None yet

## Projects

None yet

## Milestone

No milestone

## Development

Successfully merging this pull request may close these issues.

[bug] Measurement handling in convert\_qis...

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## 3 participants



fix a misused 2-qubit sample result type

d619bee

jcjaskula-aws mentioned this pull request on Dec 29, 2023

[bug] Measurement handling in convert\_qiskit\_to\_braket\_circuit sometimes fails #132

Closed

jcjaskula-aws added 2 commits 2 years ago

add test

a06b5ec

remove unnecessary import

1113521

jcjaskula-aws marked this pull request as ready for review 2 years ago

add description

d812db7

ajberdy reviewed on Dec 29, 2023

View reviewed changes

tests/providers/test\_adapter.py

Show resolved

fix test

fde7094

ajberdy approved these changes on Dec 29, 2023

View reviewed changes



rmshaffer commented on Jan 4, 2024

Collaborator

Probably orthogonal to this change, but it looks like the buggy code was originally added to fix an issue when a circuit has multiple registers: #51. Do you know if that scenario works now?

